

Review of basic Signal Processing Concepts

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Processing Discrete Signals

- Relations among sequence of samples of a signal carry information
 - Scaling all the samples by a scalar does not change these relations
 - Relations among samples are important than individual amplitudes

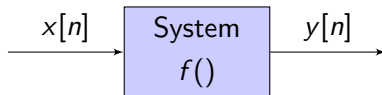
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- Process signals to extract/highlight information about the system
 - Perform mathematical operations on the sequence of samples
 - Analysis system/filter could be an algorithm or a computer program

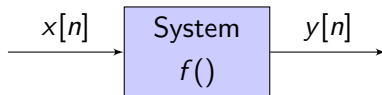
Types of Systems



- Memoryless system
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- System with memory

- Current output depends on some/all the samples around it (FIR vs IIR)

$$y[n] = f(\cdots x[n-1], x[n], x[n+1], \cdots)$$

LTI Systems



$$x_1[n] \rightarrow y_1[n]$$

$$x_2[n] \rightarrow y_2[n]$$

LTI Systems



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$$x_2[n] \rightarrow y_2[n]$$

- Linearity

- Superposition: $x_1[n] + x_2[n] \rightarrow y_1[n] + y_2[n]$
- Homogeneity: $\alpha x_1[n] \rightarrow \alpha y_1[n]$

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- Time-invariance: $x_1[n - k] \rightarrow y_1[n - k]$

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- Time-invariance: $x_1[n - k] \rightarrow y_1[n - k]$
- Linear Constant Coefficient Difference Equations

$$\sum_{k=1}^M a[k]y[n - k] = \sum_{k=1}^N b[k]x[n - k]$$

Unit Sample Response



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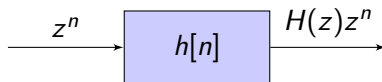
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- Response of the system for arbitrary input $x[n]$ is

$$x[n] \rightarrow y[n] = \sum_k x[k] h[n - k]$$

- Unit sample response also completely characterizes the LTI system

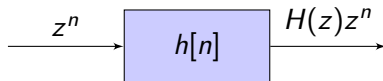
Eigen Functions of LTI System



- For inputs of the type $x[n] = z^n, z \in \mathcal{C}$, the output is given by

$$y[n] = \left(\sum_k h[k] z^{-k} \right) z^n = H(z) z^n$$

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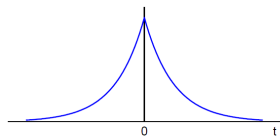
$$y[n] = \left(\sum_k h[k] z^{-k} \right) z^n = H(z) z^n$$

- For specific case of $z = e^{j\omega_0}$, the output is given by

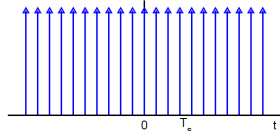
$$y[n] = \left(\sum_k h[k] e^{-j\omega_0 k} \right) e^{j\omega_0 n} = H(j\omega_0) e^{j\omega_0 n}$$

- Can we write any input as a linear combination of eigen functions?

Time-Domain Sampling

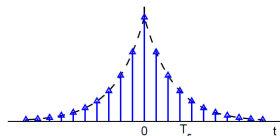


$x_c(t)$



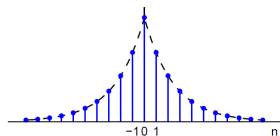
$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

$$P(j\Omega) = \frac{2\pi}{T_s} \sum_{k=-\infty}^{\infty} \delta(\Omega - k\Omega_s)$$



$$x_p(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \delta(t - nT_s)$$

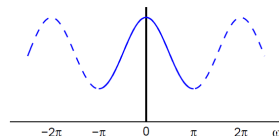
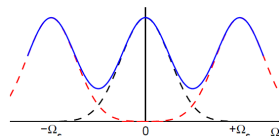
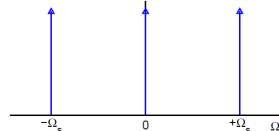
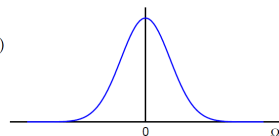
$$X_p(j\Omega) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X_c(\Omega - k\Omega_s)$$



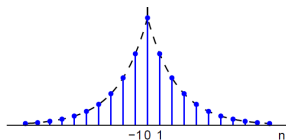
$$x[n] = x_c(nT_s)$$

$$X(j\omega) = X_p(j\Omega)|_{\Omega=\frac{\omega}{T_s}}$$

$X_c(j\Omega)$

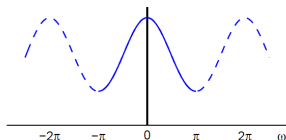


Frequency-Domain Sampling



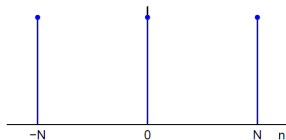
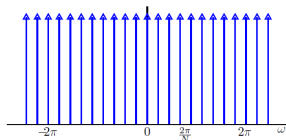
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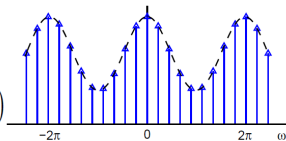
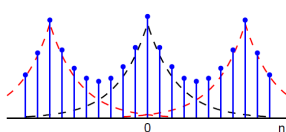
$$p_1[n] = \frac{N}{2\pi} \sum_{m=-\infty}^{\infty} \delta[n - mN]$$

$$P_1(j\omega) = \sum_{k=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi}{N}k\right)$$



$$x_{p1}[n] = \frac{N}{2\pi} \sum_{m=-\infty}^{\infty} x[n - mN]$$

$$X_{p1}(j\omega) = \sum_{k=-\infty}^{\infty} X\left(j\frac{2\pi}{N}k\right) \delta\left(\omega - \frac{2\pi}{N}k\right)$$



Fourier Transform Relations and their Connections

CTFT

$$X(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega)e^{j\Omega t} d\Omega$$

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DFT

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}kn}$$

The Z-Transform

- The Z-transform of a discrete-time signal is given by

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- Z-transform of delayed sequence: $x[n - k] \rightarrow X(z)z^{-k}$

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- LTI System described by LCCDE

$$\sum_{k=0}^M a[k]y[n - k] = \sum_{k=0}^N b[k]x[n - k]$$

- Taking Z-transform on both sides, and rearranging

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^N b[k]z^{-k}}{\sum_{k=0}^M a[k]z^{-k}}$$

Analysis of System Function

- The system function is a ratio of polynomials in z
- It can be rewritten in terms of the roots of the polynomial

$$H(z) = \frac{b[0] \prod_{k=1}^N (1 - q_k z^{-1})}{a[0] \prod_{k=1}^M (1 - p_k z^{-1})}$$

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- Poles contribute to peaks and zeros contribute to valleys of $|H(j\omega)|$
- For real systems, these roots should occur in complex conjugate pairs.
- For stable systems, all the poles should be inside the unit circle
- The contribution of a zero at $z = q$ to $|H(j\omega)|$ is same as that of a zero at $z = \frac{1}{q^*}$

Understanding Z-Plane

$$H(z) = \frac{1}{1 - az^{-1}}$$

$$h[n] = a^n u[n]$$

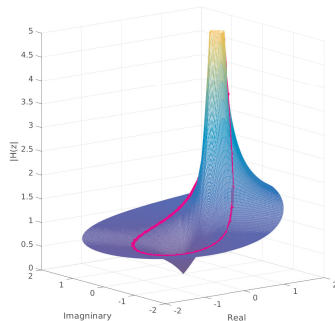
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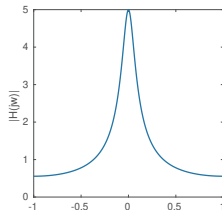
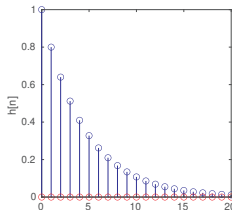
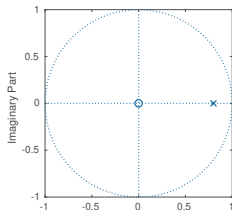
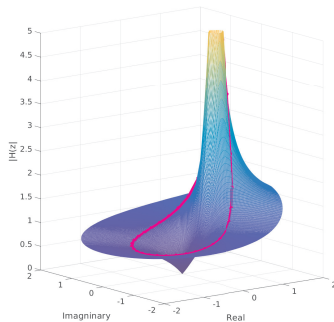


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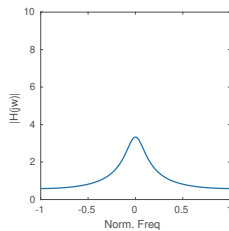
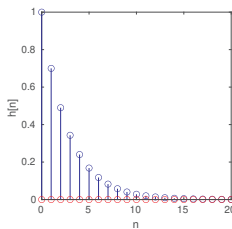
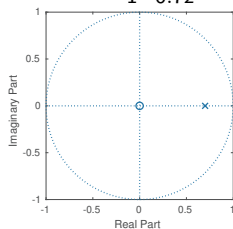
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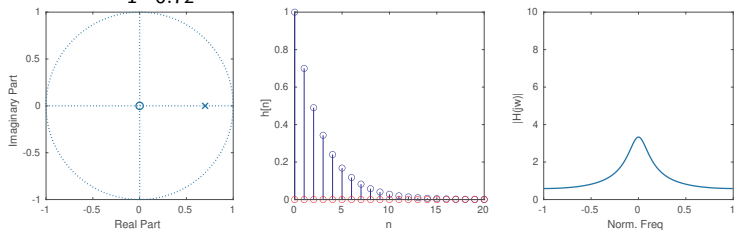
Effect of r

$H(z) = \frac{1}{1-0.7z^{-1}}$: Damping is faster, Bandwidth is higher

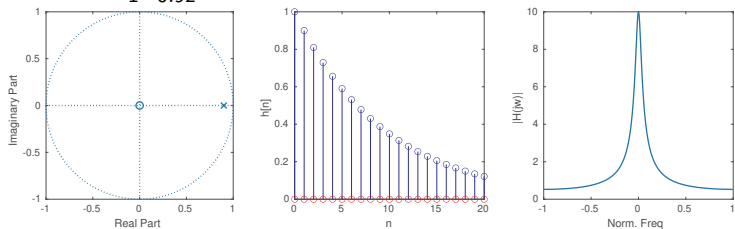


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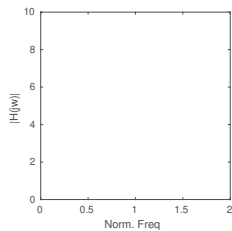
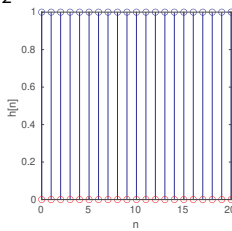
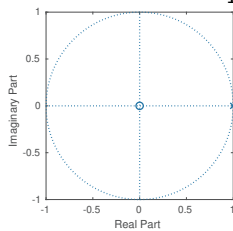


$H(z) = \frac{1}{1-0.9z^{-1}}$: Damping is slower, Bandwidth is lesser



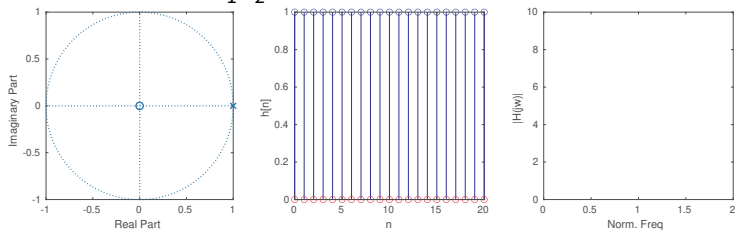
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$$H(z) = \frac{1}{1-z^{-1}}: \text{Critically-damped system}$$

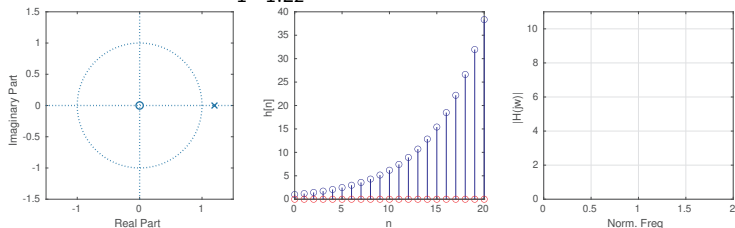


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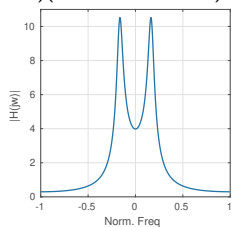
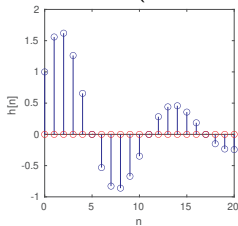
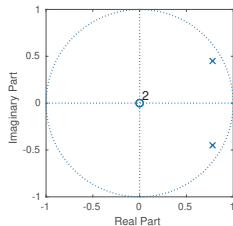


$$H(z) = \frac{1}{1-1.2z^{-1}}: \text{Over-damped system}$$



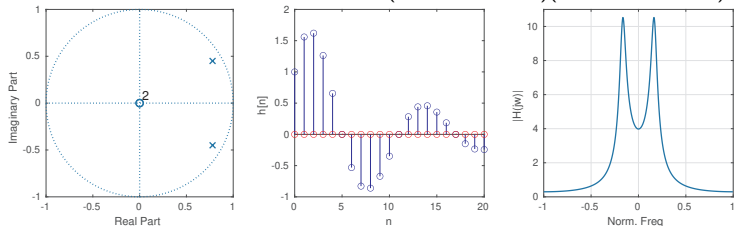
Effect of θ

$$H(z) = \frac{1}{1 - 1.5588Z^{-1} + 0.81Z^{-2}} = \frac{1}{(1 - 0.9e^{j\pi/6}Z^{-1})(1 - 0.9e^{-j\pi/6}Z^{-1})}.$$

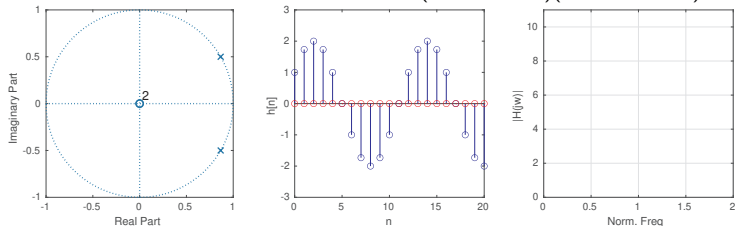


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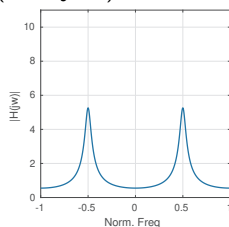
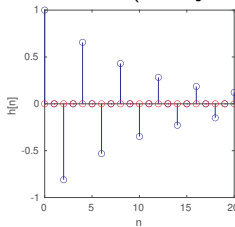
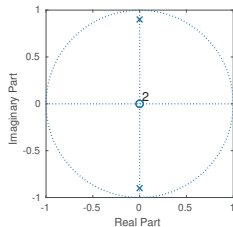


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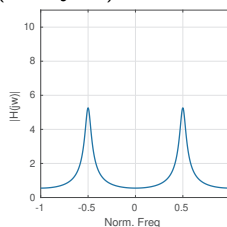
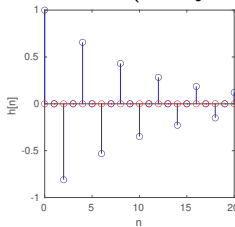
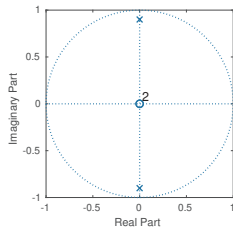
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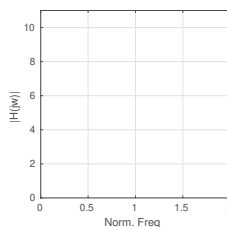
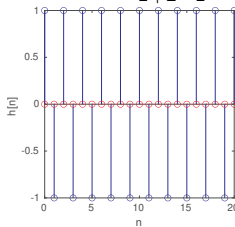
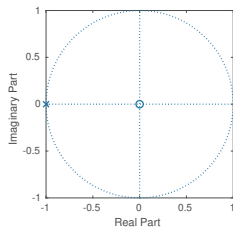


Effect of θ

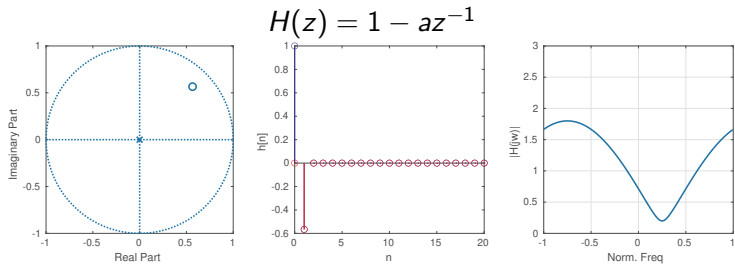
$$H(z) = \frac{1}{1-0.81z^{-2}} = \frac{1}{(1-0.9jz^{-1})(1+0.9jz^{-1})}$$



$$H(z) = \frac{1}{1+z^{-1}}$$

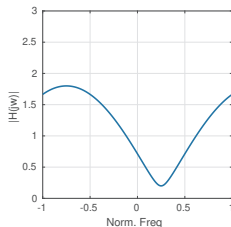
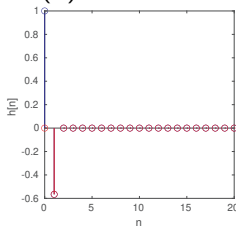
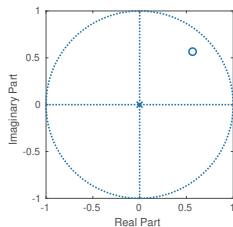


Complex Conjugate Reciprocal Zeros

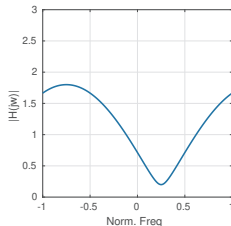
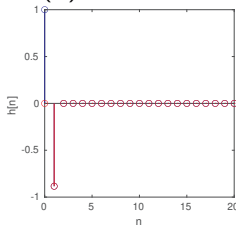
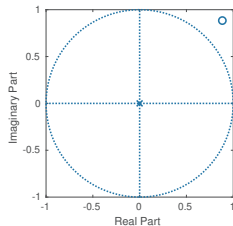


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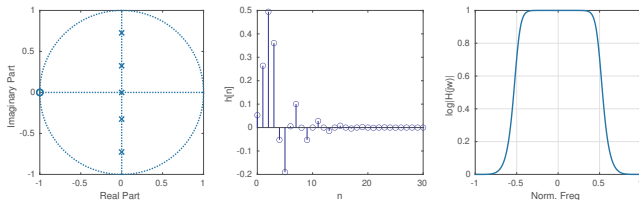
$$H(z) = 1 - az^{-1}$$



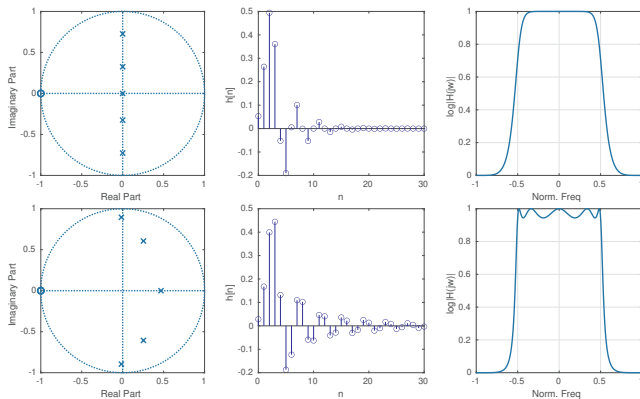
$$H(z) = z^{-1} - a^*$$



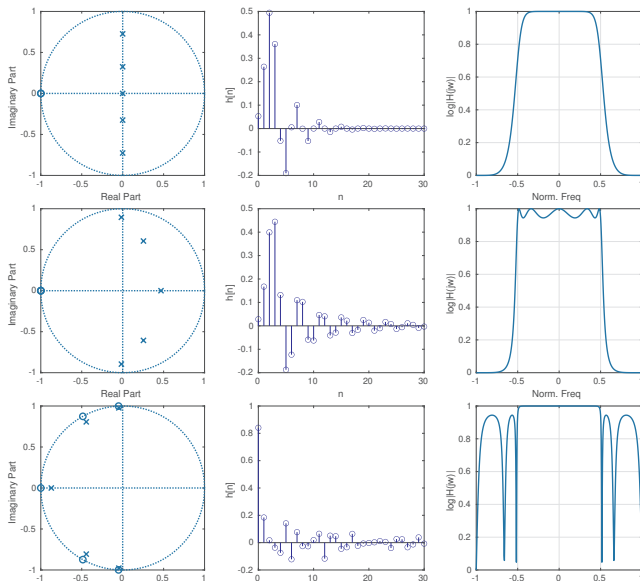
IIR Filter Design: Cutoff at $\frac{\pi}{2}$



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IIR Filter Design

- Desired filter specifications: passband gain of one, stopband gain of zero, and abrupt cutoff (zero transition band)
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- In practice, the passband and stopband exhibit ripple, and transition is gradual

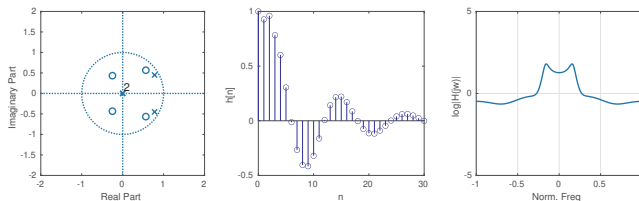
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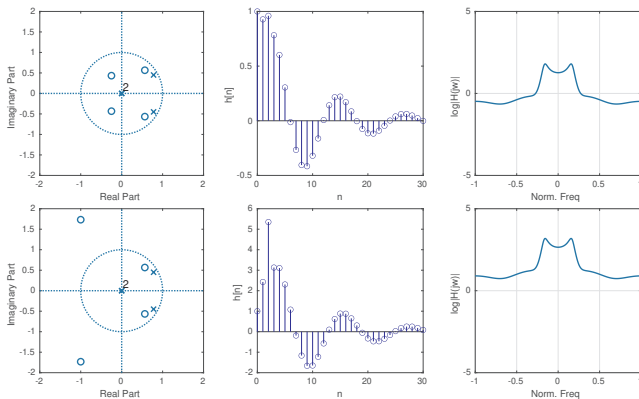
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- Most of the filter design techniques are borrowed from analog filters

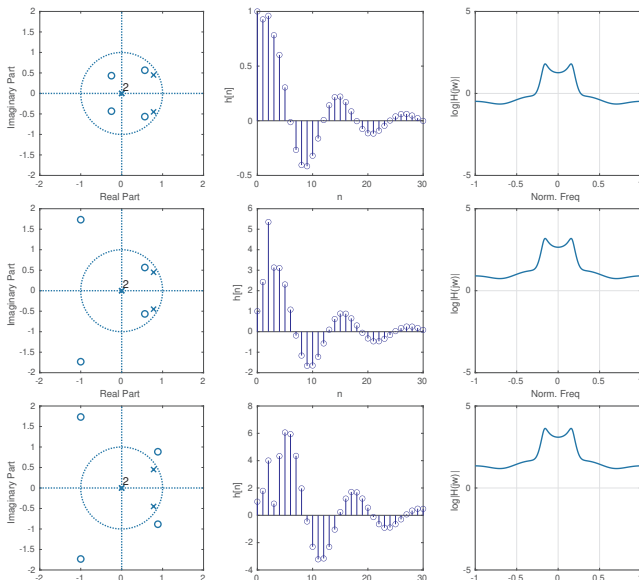
Minimum Phase Systems



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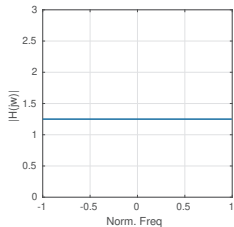
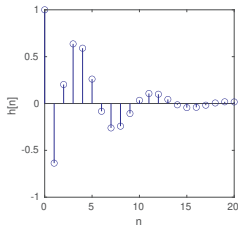
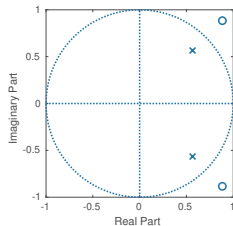


Minimum Phase Systems



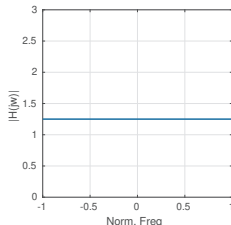
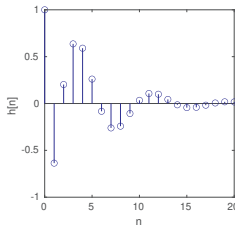
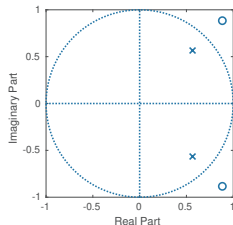
Allpass Filters

$$H(z) = \frac{z^{-1} - a^*}{1 - az^{-1}}$$



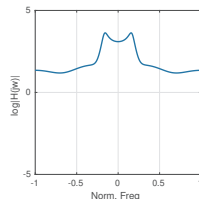
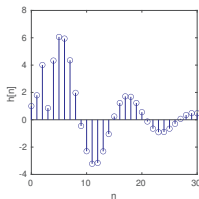
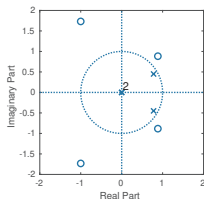
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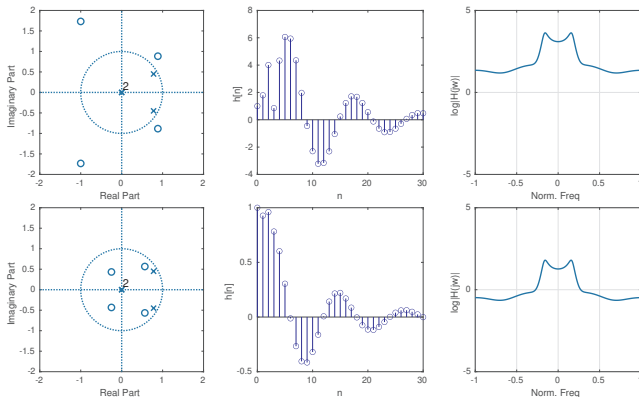


- Contribution by a pole at $z = p$ is compensated by a zero at $z = \frac{1}{p^*}$
- Poles and zeros can occur in complex conjugate reciprocal pairs.
- Useful in phase compensation, phase modeling

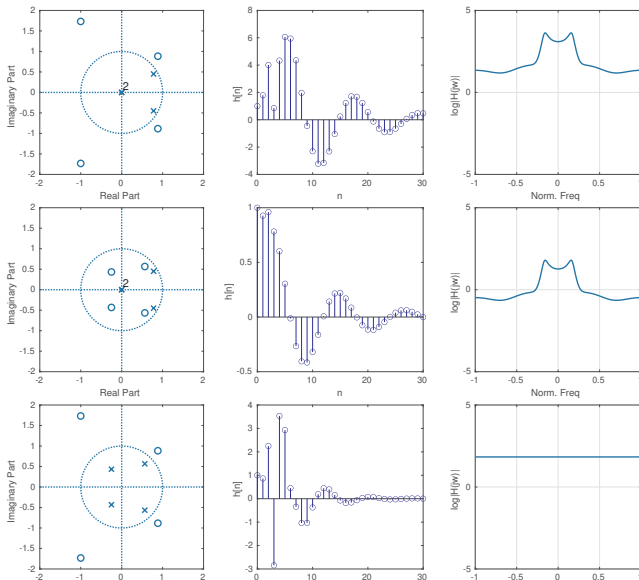
Min. Phase Allpass Decomposition: $H(z) = H_{min}(z)H_{ap}(z)$



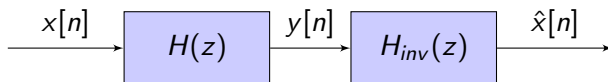
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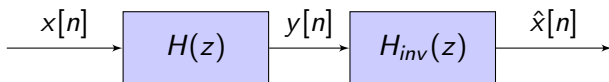


Inverse Filter



- Retrieve the input $x[n]$ from the convolved output $y[n] \rightarrow H(z)X(z)$

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- Retrieve the input $x[n]$ from the convolved output $y[n] \rightarrow H(z)X(z)$
- Design inverse filter such that $H(z)H_{inv}(z) = 1 \rightarrow h[n] * h_{inv}[n] = \delta[n]$
- The poles and zeros of $H(z)$ will be interchanged in $H_{inv}(z)$
- If $H(z)$ is min. phase, $x[n]$ can be exactly recovered $H_{inv}(z) = \frac{1}{H(z)}$
- If $H(z)$ is not min. phase, stable and causal inverse does not exist.
- If $H(z)$ is not min. phase, only magnitude response can be inverted

$$H_{inv}(z) = \frac{1}{H_{min}(z)} \leftrightarrow |H_{inv}(j\omega)| = \frac{1}{|H(j\omega)|}$$

Summary

- LTI systems can be characterized by LCCDEs, unit-sample response $h[n]$, frequency response $H(j\omega)$, or system function $H(z)$.
- Pole-zero plots of $H(z)$ helps in quickly analyzing the temporal and frequency domain characteristics of the system
- Filters can be designed with proper choice of pole-zero locations - tradeoffs have to be kept in mind
- LTI systems offer clean mathematical analysis with closed form expressions.
- Cannot handle time-varying characteristics - Adaptive filters
- Cannot handle nonlinear relationships among the samples - Kernels, RNNs

Questions?